

Topic: Union Operation

Q. Explain the Union operation in Relational Algebra. What is Union Compatibility? Provide an example to illustrate how the Union operator works.

****Definition to Remember****

The **U
into a sh
columns

Union Operation (union)

* **Binary operation** - operates on two tables.

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2. Union Compatibility (Required Condition)

For Union (and also Intersection and Set Difference) to be valid, both relations MUST be Union Compatible:

Condition	Meaning
1. Same number of attributes	Both relations must have the same number of attributes.
2. Corresponding attributes must have the same domain	For each attribute position, the domain must be the same across both relations.

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3. Example

CRICKET_TEAM

Student_ID	Name
1	Alice
2	Bob
3	Charlie
4	David
5	Eve

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FOOTBALL_TEAM

Student_ID	Name
1	Alice
2	Bob
3	Charlie
4	David
5	Eve

| :--- | :-

Query: Find all students who play Cricket OR Football or both.

CRICKET

Result: Alice, Bob, Charlie, David, Eve - 5 rows *(Charlie appeared in both tables but appears only once - duplicate eliminated).*

****Must-Write Points (for 10 marks)****

1. Union

Quick Recall

`union`
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